

Packet Tracer – VLAN Trunking Enterprise Network with VTP and Redundant Links

Objectives

Part 1: Build a Multi-Switch Enterprise Topology

Part 2: Configure VLANs

Part 3: Configure VTP

Part 4: Configure Trunk Links

Part 5: Verify Inter-Switch VLAN Communication

Part 6: Test VLAN Isolation

Part 7: Verify Redundant Links with STP

Background / Scenario

You are the network administrator of a growing company with multiple departments distributed across different floors of a building.

To simplify management:

- VLANs will separate departments.
- VTP will distribute VLAN information automatically.
- Trunk links will carry multiple VLANs between switches.
- Redundant links will provide fault tolerance.
- STP will prevent Layer 2 loops.

In this lab, you will build a small enterprise switching environment using VLANs, VTP, trunking, and redundancy.

Instructions

Part 1: Build the Topology

Step 1: Add Devices

Add:

- 3 Switches (2960)
- 4 PCs

Rename:

- Switch0 → SW1
- Switch1 → SW2
- Switch2 → SW3
- PC0 → HR-PC1
- PC1 → HR-PC2
- PC2 → SALES-PC1
- PC3 → SALES-PC2

Step 2: Connect Devices

Use Copper Straight-Through cables.

End Devices

Device	Interface	Device	Interface
HR-PC1	Fa0	SW1	Fa0/1
SALES-PC1	Fa0	SW1	Fa0/2
HR-PC2	Fa0	SW3	Fa0/1
SALES-PC2	Fa0	SW3	Fa0/2

Inter-Switch Links

Device	Interface	Device	Interface
SW1	Fa0/23	SW2	Fa0/23
SW2	Fa0/24	SW3	Fa0/24

Device Interface Device Interface

SW1 Fa0/22 SW3 Fa0/22

This creates a redundant triangle topology.

Part 2: Configure VLANs

VLAN Design

VLAN Name

10 HR

20 SALES

99 MANAGEMENT

Step 1: Configure SW1 as VTP Server

enable

configure terminal

vtp domain COMPANY

vtp mode server

vlan 10

name HR

vlan 20

name SALES

vlan 99

name MANAGEMENT

end

Step 2: Verify VLAN Creation

show vlan brief

Expected:

VLAN 10 HR

VLAN 20 SALES

VLAN 99 MANAGEMENT

Part 3: Configure VTP Clients

Configure SW2

enable

configure terminal

vtp domain COMPANY

vtp mode client

end

Configure SW3

enable

configure terminal

vtp domain COMPANY

vtp mode client

end

Verify VTP

On SW2:

show vtp status

Expected:

VTP Operating Mode: Client

Domain Name: COMPANY

Part 4: Configure Trunk Links

Step 1: Configure Trunks on SW1

configure terminal

interface range fa0/22 - 23

switchport mode trunk

switchport trunk allowed vlan 10,20,99

end

Step 2: Configure Trunks on SW2

configure terminal

interface range fa0/23 - 24

switchport mode trunk

switchport trunk allowed vlan 10,20,99

end

Step 3: Configure Trunks on SW3

configure terminal

interface range fa0/22 , fa0/24

switchport mode trunk

switchport trunk allowed vlan 10,20,99

end

Verify Trunks

show interfaces trunk

Expected:

Mode: trunk

Allowed VLANs: 10,20,99

Part 5: Assign Access Ports

SW1

configure terminal

interface fa0/1

switchport mode access

switchport access vlan 10

interface fa0/2

switchport mode access

switchport access vlan 20

end

SW3

configure terminal

interface fa0/1

switchport mode access

switchport access vlan 10

interface fa0/2

switchport mode access

switchport access vlan 20

end

Verify VLAN Membership

show vlan brief

Expected:

Fa0/1 VLAN 10

Fa0/2 VLAN 20

Part 6: Configure PC IP Addresses

VLAN 10 – HR

HR-PC1

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Gateway: Blank

HR-PC2

IP Address: 192.168.10.20

Subnet Mask: 255.255.255.0

Gateway: Blank

VLAN 20 – SALES

SALES-PC1

IP Address: 192.168.20.10

Subnet Mask: 255.255.255.0

Gateway: Blank

SALES-PC2

IP Address: 192.168.20.20

Subnet Mask: 255.255.255.0

Gateway: Blank

Part 7: Verify VLAN Communication

Test 1: HR VLAN Communication

From HR-PC1:

ping 192.168.10.20

Expected:

Reply from 192.168.10.20

4 successful replies.

Test 2: SALES VLAN Communication

From SALES-PC1:

ping 192.168.20.20

Expected:

Reply from 192.168.20.20

4 successful replies.

Test 3: VLAN Isolation

From HR-PC1:

ping 192.168.20.20

Expected:

Request Timed Out

Without a router or Layer 3 switch, VLANs cannot communicate.

Part 8: Verify STP Redundancy

Because the switches form a triangle topology:

SW1
/ \
/ \
SW2-----SW3

STP should block one redundant path.

Verify STP

show spanning-tree

Expected:

One port in Blocking State

or

Alternate Port

depending on Packet Tracer version.

Verify Root Bridge

show spanning-tree

Expected:

This bridge is the root

on one switch.

Part 9: Test Redundant Link Failover

Disconnect:

SW1 Fa0/23

to

SW2 Fa0/23

Wait 30–50 seconds.

Verify:

ping 192.168.10.20

Expected:

Reply received

Traffic should automatically use the backup path.

Verification Commands

Verify VLANs

show vlan brief

Verify Trunks

show interfaces trunk

Verify VTP

show vtp status

Verify STP

show spanning-tree

Verify MAC Learning

show mac address-table

Troubleshooting Tips

If VLAN communication fails:

✓ Verify VLANs exist

show vlan brief

✓ Verify trunk links

show interfaces trunk

✓ Verify VTP domain name matches

show vtp status

✓ Verify access port assignments

✓ Verify PCs are connected to correct VLAN ports

✓ Verify trunk links allow VLANs 10,20,99

✓ Wait for STP convergence

✓ Verify one trunk is not accidentally shutdown

What You Learned

- How VLANs segment a network
 - How VTP distributes VLAN information
 - How trunk ports carry multiple VLANs
 - How access ports assign devices to VLANs
 - How VLAN isolation works
 - How STP prevents Layer 2 loops
 - How redundant links provide fault tolerance
 - How to verify enterprise switching configurations
-

Challenge Extensions (Optional)

Challenge 1: Add an IT VLAN

Create:

VLAN 30

Name: IT

Add two PCs and test communication.

Challenge 2: Configure a VTP Password

On all switches:

ntp password Cisco123

Verify synchronization still works.

Challenge 3: Add Router-on-a-Stick

Add a router and configure:

Inter-VLAN Routing

Verify HR and SALES VLANs can communicate.

Challenge 4: Change Root Bridge

Force SW2 to become Root Bridge:

spanning-tree vlan 1 root primary

Observe STP recalculation.

Challenge 5: Use Simulation Mode

Switch Packet Tracer to:

Simulation Mode

Observe:

- VTP Advertisements
- STP BPDUs
- ARP Requests
- ICMP Packets

and follow their path through the network.